

Observable Greenhouse Trainer

Observe, decide, actuate, present, record

A deterministic project with visible evidence and no energetic load

Lesson	024 — project	Time	45–75 minutes
Prerequisites	Lessons 001–023	Board	Arduino Mega 2560
Evidence	TP-M, D38, D13, RGB, LCD	Status	host verified
Energy class	E1: USB, passive input, LCD, and resistor-limited LEDs only		

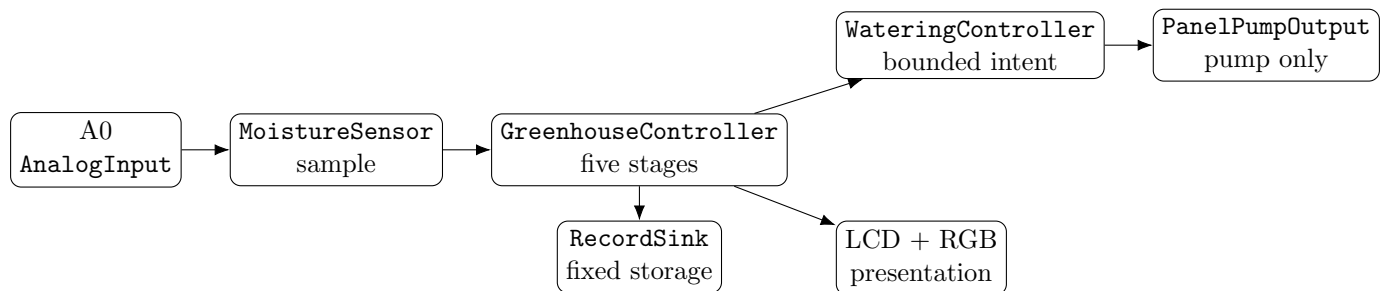
Hard boundary. “Pump” means the LED on D38. Do not connect a relay, motor, pump, heater, water, external load supply, or mains circuit. The fixed storage medium is an in-memory teaching model, not an SD card. ADK is not an unattended irrigation or safety controller.

1 Mission and dependency map

Build one replayable chain:

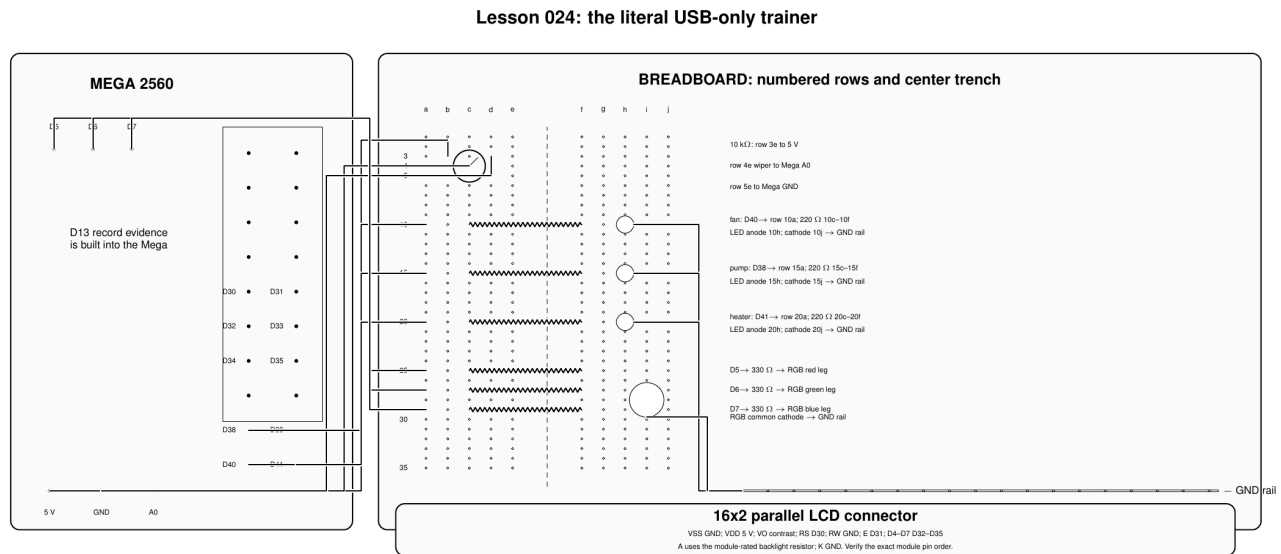
TP-M → MoistureSensor → WateringController → PanelPumpOutput → InertLoadPanel

↘ LCD view + stable record + RGB mode evidence.



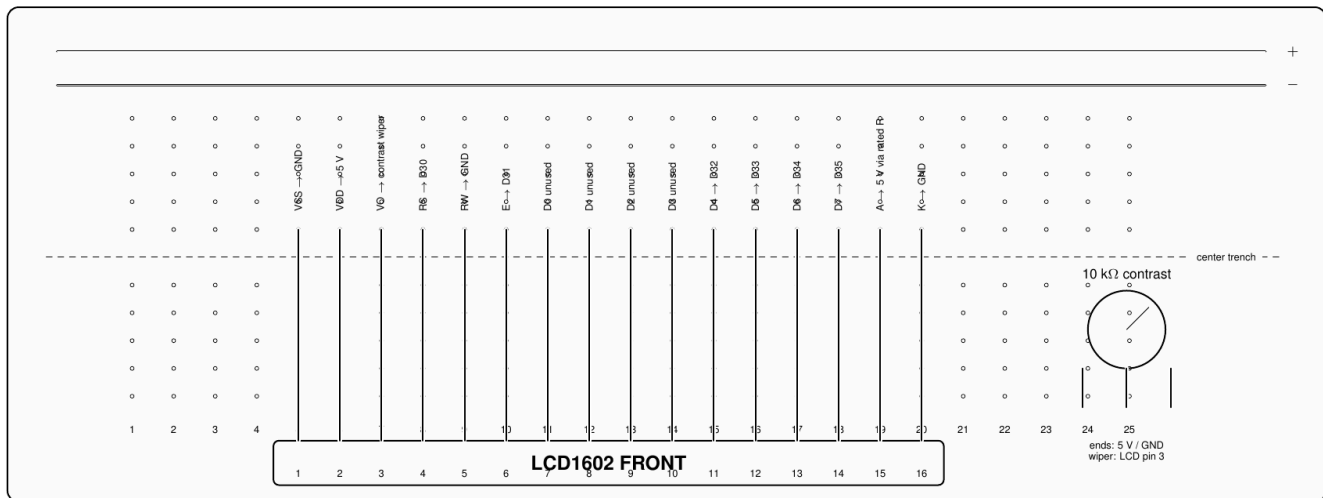
The arrows are ownership or explicit borrowing relationships, not hidden global access. The coordinator owns dependency lifecycle calls; caller-owned objects outlive it.

2 Orientation and exact circuit



Alternative text: literal Mega and numbered-breadboard plate. It shows the paired D30–D41 header, top-header D5–D7, A0, 5 V, and GND; exact potentiometer rows; three separate resistor–LED–ground paths; three RGB series paths; and the parallel LCD connector. D13 is explicitly the built-in LED. The plate contains no relay or external load.

LCD close-up: header pins 1–16 occupy breadboard F5–F20



Pins 7–10 stay empty. If the module's printed order differs, do not power it.

Alternative text: close-up of all sixteen LCD header positions, numbered 1 through 16 in breadboard holes F5 through F20. Every used supply, contrast, control, four-bit data, and backlight connection is labeled; D0 through D3 remain empty.

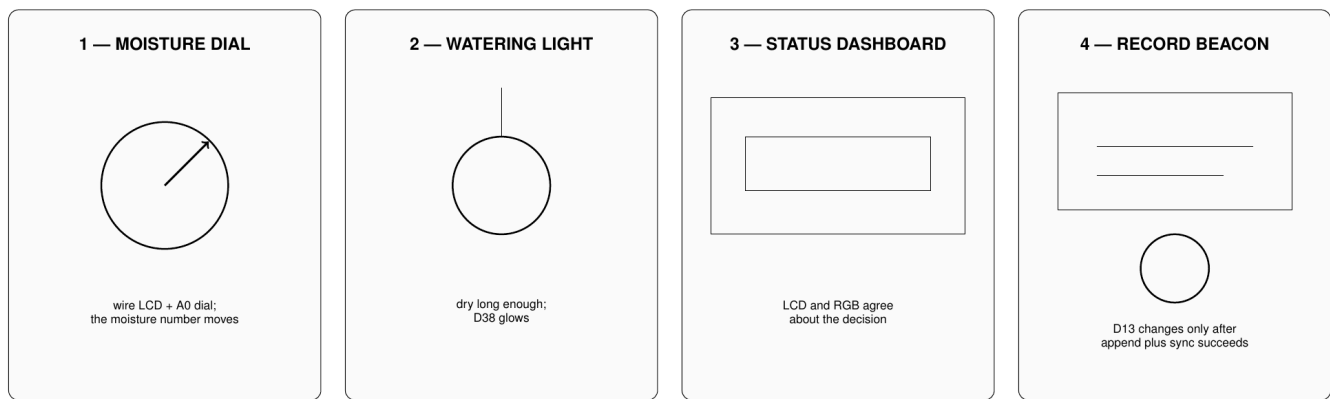
The drawing helps locate parts; this table is authoritative.

Meaning	Pin	Connection
Moisture simulator / TP-M	A0 / D54	10 k Ω potentiometer wiper
Potentiometer ends	5 V, GND	one end to each USB-powered rail
Simulated fan	D40	220 Ω , LED anode; cathode to GND
Simulated pump / TP38	D38	220 Ω , LED anode; cathode to GND
Simulated heater	D41	220 Ω , LED anode; cathode to GND
Record acknowledgement	D13	built-in LED
Mode evidence	D5, D6, D7	RGB channels, one 330 Ω resistor each
LCD control/data	D30–D35	RS, E, D4, D5, D6, D7
LCD supply	5 V, GND	exact module rating and contrast network

Before USB: trace each signal from its labeled Mega pin, through one series resistor and LED, to GND. Check the LCD's printed pin order against the table. If the module differs, stop at the three-LED build and leave the LCD unplugged. There is no relay or external supply in this lesson.

3 Four quick wins

Four quick builds; each reward depends on the previous one



All outputs are LEDs. After 120 seconds the sketch shuts every owned circuit down.

Alternative text: four panels show the dependent progression: potentiometer-controlled moisture number, dry-triggered D38 watering LED, LCD-and-RGB status dashboard, and D13 acknowledgement after an in-memory record append and sync.

1. Wire the potentiometer and LCD. Turning the dial must move the moisture value; if it does not, later stages cannot make a valid decision.
2. Add D38 and its resistor. Hold the dial in the dry region: after the minimum-off interval D38 lights. Turn wet and it goes dark.
3. Add the RGB channels. Their pattern and the LCD words must explain the same D38 decision. Every eight seconds inhibition creates an amber, D38-dark intermission without another input part.
4. Watch built-in D13. It changes only after a complete record is appended and synchronized in the fixed RAM model.

At 120 seconds the sketch shuts down the controller and every owned evidence output. Press Reset for another run.

4 The narrative loop

```

const adk::TimePoint now (millis ());

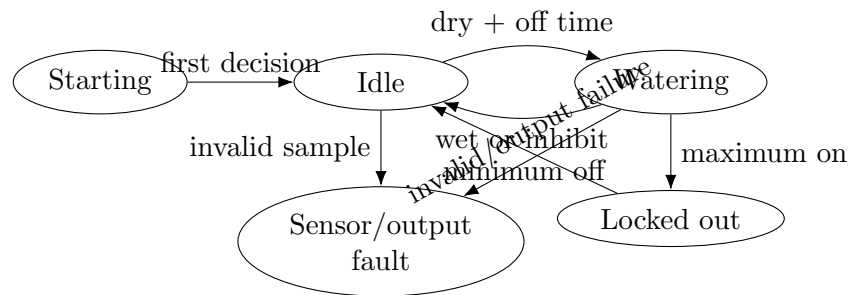
observeGreenhouse (now);
decideGreenhouse  (now);
actuateGreenhouse (now);
presentGreenhouse (now);
recordGreenhouse  (now);
showGreenhouseHealth(now);

```

observe() samples at most once when due. **decide()** changes policy and intent but no output. **actuate()** is the only stage allowed to change D38. **present()** writes an already-decided LCD view. **record()** publishes the same decision. Missed periods coalesce; they do not create a burst.

5 Watering policy and state graph

The reference calibration maps raw 800 to dry (0 permille) and raw 200 to wet (1000 permille). Watering may begin below 350 only after 2000 ms off. It stops at or above 600, on inhibition, on invalid or stale sensing, or after 5000 ms on. Maximum-on creates a lockout; another dry sample cannot clear it early. **PanelPumpOutput** maps on to the panel's pump channel and off to none. The fan and heater indicators remain off throughout this project.



Condition	Decision	D38
Invalid, unavailable, or stale sample	sensor fault dominates	off
Operator inhibition	intent forced off	off
On-time at or beyond maximum	lockout	off
Valid sample at or above wet threshold	stop watering	off
Valid sample below dry threshold after off-time	begin watering	on
Between thresholds	preserve safe current state	unchanged

6 Mode precedence and independent evidence

Mode	RGB pattern	Narrow interpretation
Starting	slow blue pulse	lifecycle/schedule start
Monitoring	green heartbeat	valid sample; pump intent off
Watering	cyan heartbeat	D38 requested on
Inhibited	steady amber	software policy denied watering
SensorFault	two red pulses	sample unusable
OutputFault	steady red	command failure latched
DisplayFault	three amber pulses	LCD cannot report itself
RecordFault	four red pulses	append or sync failed
MultipleFaults	steady red	output plus another fault

Output failure dominates and latches until successful reinitialization. Sensor failure and inhibition always suppress watering. A display or record failure cannot enable D38.

6.1 Read the visible story

1. **Acquisition.** LCD startup and the first RGB transition show that the visible system acquired its resources.
2. **Safe state.** D38 stays dark at startup, inhibition, invalid sensing, and shutdown.
3. **Record boundary.** D13 changes only after append and sync succeed. It means the RAM model accepted a record, never SD durability.

If a later reward appears without the earlier one, remove USB and retrace that stage's numbered rows. Serial is not needed to enjoy or diagnose the build.

7 What the visible sequence means

ms	raw	allowed	visible result	D38	record
0	500	yes	monitoring	off	first accepted
1999	700	yes	dry; waiting	off	cadence continues
2000	700	yes	watering	on	decision captured
4500	440	yes	wet threshold	off	stop captured
6500	700	yes	watering may restart	on	decision captured
8000	700	no	amber inhibited	off	inhibition captured
16000	700	yes	policy resumes	timing decides	cadence continues

The repository's automated tests replay exact timestamp and fault traces behind the scenes while the learner experience stays with the visible dependent sequence.

8 Stable records, storage, and time

Version 1 uses bounded locale-independent ASCII:

```
adk-gh,1,17,42000,valid,472,340,on,dry,watering
```

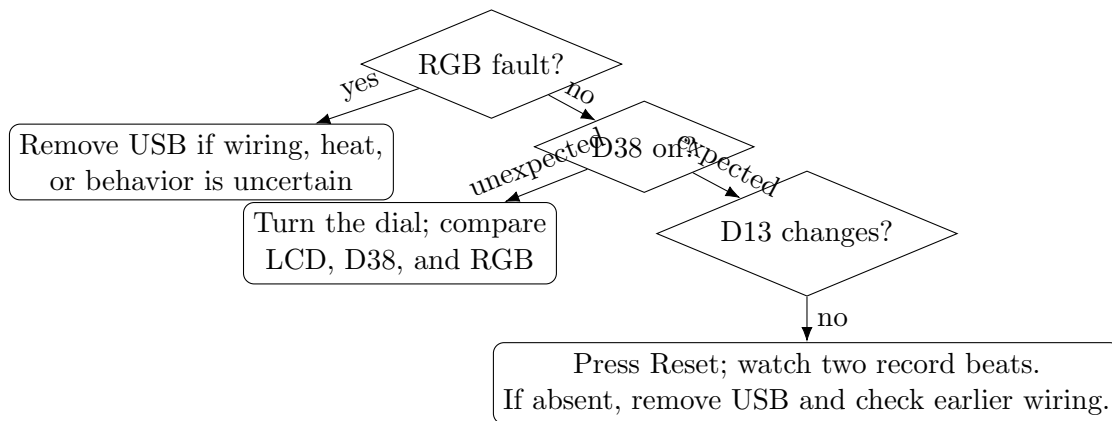
Fields are format, version, sequence, decision milliseconds, sample state, raw reading, moisture permille or a dash, requested pump, reason, and mode. Sequence advances only after successful acceptance. One failed pending record keeps its original timestamp and bytes during retry.

FixedStorage distinguishes staged bytes from a durable prefix: `append()` alone is not durability; `sync()` advances the durable prefix. Shutdown and restart discard an unsynchronized suffix. This is an executable storage model in RAM, not evidence for an SD card.

The public **Rtc** boundary distinguishes valid, unset, oscillator-stopped, and transport-fault readings. The deterministic **SimulatedRtc** returns a fixed epoch plus a sequence. The record sink requires a valid reading before append and sync. No physical RTC adapter is selected, so the record schema still uses explicit **TimePoint** milliseconds.

9 Fault matrix and diagnosis

Injected condition	Required response	Evidence to retain
Calibration invalid	initialization rejects, no output claim	complete rollback
Range or stale sample	D38 off; sensor fault visible	raw reading and state
Output command failure	best-effort off; latched output fault	full status
LCD write failure	watering state unchanged	RGB display-fault pattern
Append failure	pending record retained	original bytes and sequence
Sync failure	no durability claim; retry same record	D13 does not acknowledge
Delayed stage call	one coalesced operation	timestamp and next snapshot
Shutdown in any mode	dependencies close in reverse	TP38 safe state



Never diagnose by shorting, hot-plugging bare wires, or attaching a real load.

10 Safe finish and honest claims

The last reward is the shutdown: at 120 seconds D38, the other inert-load LEDs, the RGB indicator, display, record sink, moisture input, and their resource claims all close in dependency-safe order. Remove USB before changing wires.

An LED and fixed-storage model cannot establish pump electrical behavior, SD durability, RTC accuracy, water delivery, plant health, or unattended safety. No physical RTC, media adapter, pump, relay, heater, or water path is part of this build, and no bench verification is claimed by this document.

11 References and next step

- Canonical Lesson 024 sketch

- Controller interface and deterministic tests
- Lessons 022–024 design record and ADK safety model
- Arduino Mega 2560 Rev3 documentation
- Arduino data-logging guide

The HTML reference carries searchable API links. This PDF is the printable build adventure. Next, Lesson 025 separates captured infrared timing evidence from decoded command policy.